Statistical Signal Processing

Laboratory 2:
Bayesian filtering

1 Filtering for a linear-Gaussian state space model

1.1 Model

We tackle the problem of tracking an object moving on a plane. The motion dynamics can be described by the pair of equations

$$\mathbf{v}_n = \mathbf{v}_{n-1} + \mathbf{u}_n, \tag{1}$$

$$\mathbf{r}_n = \mathbf{r}_{n-1} + T\mathbf{v}_{n-1} + \mathbf{w}_n, \tag{2}$$

where, at time n, $\mathbf{v}_n$ is the 2×1 velocity vector, $\mathbf{r}_n$ is the position vector, $\mathbf{u}_n \sim \mathcal{N}(\mathbf{u}_n; \mathbf{0}, \sigma_u^2 \mathbf{I}_2)$ and $\mathbf{w}_n \sim \mathcal{N}(\mathbf{w}_n; \mathbf{0}, \sigma_w^2 \mathbf{I}_2)$ are Gaussian noise vectors with diagonal covariance matrices

$$\sigma_u^2 \mathbf{I}_2 = \begin{bmatrix} \sigma_u^2 & 0 \\ 0 & \sigma_u^2 \end{bmatrix} \quad \text{and} \quad \sigma_w^2 \mathbf{I}_2 = \begin{bmatrix} \sigma_w^2 & 0 \\ 0 & \sigma_w^2 \end{bmatrix},$$

and $T > 0$ is the length of the time step.

The model in Eqs. (1) and (2) can be compactly represented as

$$\mathbf{x}_n = \mathbf{A}\mathbf{x}_{n-1} + \mathbf{s}_n, \tag{3}$$

where

$$\mathbf{x}_n = \begin{bmatrix} \mathbf{r}_n \\ \mathbf{v}_n \end{bmatrix}$$

is the 4×1 state vector,

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

is the transition matrix, and $\mathbf{s}_n \sim \mathcal{N}(\mathbf{s}_n; \mathbf{0}, \mathbf{Q})$ is zero-mean Gaussian noise with covariance matrix

$$\mathbf{Q} = \begin{bmatrix} \sigma_w^2 & 0 & 0 & 0 \\ 0 & \sigma_w^2 & 0 & 0 \\ 0 & 0 & \sigma_u^2 & 0 \\ 0 & 0 & 0 & \sigma_u^2 \end{bmatrix}.$$

At each time step, we have linear observations of the form

$$\mathbf{y}_n = \mathbf{H}\mathbf{x}_n + \mathbf{l}_n,$$

where

$$\mathbf{H} = \begin{bmatrix} 1 & -0.3 & 0 & 0 \\ -0.2 & -1 & 0 & 0 \end{bmatrix}$$

is the observation matrix and $\mathbf{l}_n \sim \mathcal{N}(\mathbf{l}_n; \mathbf{0}, \sigma_l^2 \mathbf{I}_2)$ is Gaussian noise. Note that the observations are 2×1 vectors that depend on the position of the state $\mathbf{r}_n$ alone (and not on its velocity $\mathbf{v}_n$).

The initial probability distribution for the state is Gaussian, $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{x}_0; \mathbf{0}, \mathbf{P}_0)$, where $\mathbf{P}_0 = \sigma_0^2 \mathbf{I}_4$.

1.2 Exercises

1.2.1 Simulation of the model

Code a program to simulate the state space model in Section 1 for $T = \frac{1}{2}$, $\sigma_0^2 = \frac{1}{5}$, $\sigma_u^2 = \frac{1}{10}$, $\sigma_w^2 = \frac{1}{2}$ and $\sigma_l^2 = 4$. If the vectors `x` and `y` contain the sequences of states and linear observations, respectively, plot:

1. Three sample trajectories, using

   $$\texttt{plot}(\texttt{x}(1, 1:\texttt{end}), \texttt{x}(2, 1:\texttt{end}), '\texttt{k}-')$$

   for each simulation. Plot the velocities versus time, component-wise, as

   $$\texttt{plot}(1:\texttt{m}+1, \texttt{x}(3, 1:\texttt{m}+1), '\texttt{k}-');$$
   $$\texttt{plot}(1:\texttt{m}+1, \texttt{x}(4, 1:\texttt{m}+1), '\texttt{r}-');$$

2. The linear observations for the three simulations, using

   $$\texttt{plot}(1:\texttt{m}+1, \texttt{y}(1, 1:\texttt{m}+1), '\texttt{b}-')$$

   and

   $$\texttt{plot}(1:\texttt{m}+1, \texttt{y}(2, 1:\texttt{m}+1), '\texttt{r}-').$$

1.2.2 Kalman filter

Assume there are $m = 100$ observations, $\mathbf{y}_1, \ldots, \mathbf{y}_m$, and the parameter values are $T = \frac{1}{2}$, $\sigma_0^2 = \frac{1}{5}$, $\sigma_u^2 = \frac{1}{10}$, $\sigma_w^2 = \frac{1}{2}$ and $\sigma_l^2 = 4$. Use the Kalman filter to compute the sequence of probability density functions (pdf's)

$$p(\mathbf{x}_n|\mathbf{y}_{1:n}) = \mathcal{N}(\mathbf{x}_n; \hat{\mathbf{x}}_n, \mathbf{P}_n), \quad n = 1, \ldots, m,$$

where $\hat{\mathbf{x}}_n = \mathbb{E}[\mathbf{x}_n|\mathbf{y}_{1:n}]$ is the conditional mean and

$$\mathbf{P}_n = \mathbb{E}[(\mathbf{x}_n - \hat{\mathbf{x}}_n)(\mathbf{x}_n - \hat{\mathbf{x}}_n)^\top|\mathbf{y}_{1:n}]$$

is the conditional covariance matrix of $\mathbf{x}_n$ given the observations $\mathbf{y}_{1:n}$. Note that the state means have the form

$$\hat{\mathbf{x}}_n = \begin{bmatrix} \hat{\mathbf{r}}_n \\ \hat{\mathbf{v}}_n \end{bmatrix}.$$

Plot, for three independent simulations:

1. The trajectory $\mathbf{r}_0, \mathbf{r}_1, \ldots, \mathbf{r}_m$ and the estimates $\hat{\mathbf{r}}_0, \hat{\mathbf{r}}_1, \ldots, \hat{\mathbf{r}}_m$.
2. The velocities $\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_m$ and the estimates $\hat{\mathbf{v}}_0, \hat{\mathbf{v}}_1, \ldots, \hat{\mathbf{v}}_m$.
3. The mean square errors $\mathrm{MSE}_r(n) := \mathbb{E}[\|\mathbf{r}_n - \hat{\mathbf{r}}_n\|^2]$ and $\mathrm{MSE}_v(n) := \mathbb{E}[\|\mathbf{v}_n - \hat{\mathbf{v}}_n\|^2]$ computed using the covariance matrix $\mathbf{P}_n$, for $n = 0, \ldots, m$.

Repeat the exercise with an increased noise variance, $\sigma_l^2 = 16$.

1.2.3 Particle filter

For the same setup as in Section 1.2.2, use the standard particle filter (PF) to *approximate* the sequence of probability distributions $p(\mathbf{x}_n|\mathbf{y}_{1:n})$, for $n = 1, \ldots, m$.

The standard PF with N particles can be outlined as follows.

1. **Initialisation**: generate N i.i.d. samples from the prior pdf

   $$p(\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0; \mathbf{0}, \sigma_0^2 \mathbf{I}_4),$$

   denoted $\mathbf{x}_0^1, \ldots, \mathbf{x}_0^N$, where $\mathbf{x}_0^i = \begin{bmatrix} \mathbf{r}_0^i \\ \mathbf{v}_0^i \end{bmatrix}$.

2. **Recursive step**: assume that we have particles $\mathbf{v}_{n-1}^i$ and $\mathbf{r}_{n-1}^i$, for $i = 1, \ldots, N$, generated at time $n-1$. Then, at time n we take the following steps:

(a) Sample new particles: for $i = 1, \ldots, N$, draw

$$\mathbf{u}_n^i \sim \mathcal{N}(\mathbf{u}_n; \mathbf{0}, \sigma_u^2 \mathbf{I}_2) \quad \text{and} \quad \mathbf{w}_n^i \sim \mathcal{N}(\mathbf{u}_n; \mathbf{0}, \sigma_w^2 \mathbf{I}_2),$$

and then compute $\bar{\mathbf{x}}_n^i = \begin{bmatrix} \bar{\mathbf{r}}_n^i \\ \bar{\mathbf{v}}_n^i \end{bmatrix}$, where

$$\begin{aligned} \bar{\mathbf{v}}_n^i &= \mathbf{v}_{n-1}^i + \mathbf{u}_n^i, \\ \bar{\mathbf{r}}_n^i &= \mathbf{r}_{n-1}^i + T\mathbf{v}_{n-1}^i + \mathbf{w}_n^i. \end{aligned}$$

(b) Compute weights: the normalized weights in the standard particle filter have the form

$$\omega_n^i = \frac{p(\mathbf{y}_n|\bar{\mathbf{x}}_n^i)}{\sum_{j=1}^N p(\mathbf{y}_n|\bar{\mathbf{x}}_n^j)}. \tag{4}$$

A problem with this expression is that

$$p(\mathbf{y}_n|\bar{\mathbf{x}}_n^i) \propto \mathsf{e}^{-\frac{1}{2\sigma_l^2}\|\mathbf{y}_n - \mathbf{H}\bar{\mathbf{x}}_n^i\|^2},$$

and the exponential function can take values which are 0 up to the machine precision. Then, (4) can yield a division by 0.
To avoid this, we work with the logarithm of the weights. Specifically, for $i = 1, \ldots, N$ we compute the log-likelihoods

$$\lambda_n^i = -\frac{1}{2\sigma_l^2}\|\mathbf{y}_n - \mathbf{H}\bar{\mathbf{x}}_n^i\|^2$$

and the maximum log-likelihood

$$\Lambda_n = \max_i \lambda_n^i.$$

The non-normalized importance weights are then calculated as

$$\tilde{\omega}_n^i = \mathsf{e}^{\lambda_n^i - \Lambda_n}, \quad \text{for } i = 1, ..., N,$$

and the normalized weights are

$$\omega_n^i = \frac{\tilde{\omega}_n^i}{\sum_{j=1}^N \tilde{\omega}_n^j}.$$

Note that there is at least one weight of the form $\tilde{\omega}_n^i = \mathsf{e}^0 = 1$ and hence $\sum_{j=1}^N \tilde{\omega}_n^j \geq 1$.

(c) Resampling: for $i = 1, \ldots, N$, set $\mathbf{x}_n^i = \bar{\mathbf{x}}_n^j$ with probability ω_n^j, $j \in \{1, \ldots, N\}$.

In Matlab, this is implemented easily using the randsample function. Assume that we have the particles stored in a $4 \times N$ matrix X, where the i-th column of X represents the particle $\bar{\mathbf{x}}_n^i$, and the normalized weights are stored in an $N \times 1$ vector W. Then we generate N i.i.d. random indices in $\{1, \ldots, N\}$ with probabilities given by the weights with the command

$$\mathsf{idx} = \mathsf{randsample}(1 : \mathsf{N}, \mathsf{N}, \mathsf{true}, \mathsf{W})$$

and then assign the resampled particles with

$$\begin{aligned} \mathsf{Xold} &= \mathsf{X}; \\ \mathsf{X}(:, 1 : \mathsf{N}) &= \mathsf{Xold}(:, \mathsf{idx}). \end{aligned}$$

(d) Estimate the state $\mathbf{x}_n$ as

$$\hat{\mathbf{x}}_n^N = \sum_{i=1}^{N} \omega_n^i \bar{\mathbf{x}}_n^i.$$

Set $N = 100$ and plot, for three independent simulations:

1. The trajectory $\mathbf{r}_0, \mathbf{r}_1, \ldots, \mathbf{r}_m$, the estimates $\hat{\mathbf{r}}_0, \hat{\mathbf{r}}_1, \ldots, \hat{\mathbf{r}}_m$ (from the Kalman filter) and the estimates $\hat{\mathbf{r}}_0^N, \hat{\mathbf{r}}_1^N, \ldots, \hat{\mathbf{r}}_m^N$ (from the PF).

2. The velocities $\mathbf{v}_0, \mathbf{v}_1, \ldots, \mathbf{v}_m$, the estimates $\hat{\mathbf{v}}_0, \hat{\mathbf{v}}_1, \ldots, \hat{\mathbf{v}}_m$ (from the Kalman filter) and the estimates $\hat{\mathbf{v}}_0^N, \hat{\mathbf{v}}_1^N, \ldots, \hat{\mathbf{v}}_m^N$ (from the PF).

3. The square errors

$$\mathcal{E}_r^{KF}(n) := \|\mathbf{r}_n - \hat{\mathbf{r}}_n\|^2 \quad \text{and} \quad \mathcal{E}_v^{KF}(n) := \|\mathbf{v}_n - \hat{\mathbf{v}}_n\|^2,$$

from the Kalman filter, and

$$\mathcal{E}_r^{PF}(n) := \|\mathbf{r}_n - \hat{\mathbf{r}}_n^N\|^2 \quad \text{and} \quad \mathcal{E}_v^{PF}(n) := \|\mathbf{v}_n - \hat{\mathbf{v}}_n^N\|^2,$$

from the PF, versus time $n = 1, \ldots, m$.

Repeat the exercise with an increased noise variance, $\sigma_l^2 = 16$. Do you need to increase N?

2 A stochastic FitzHugh-Nagumo model

2.1 Model

The FitzHugh-Nagumo system is a simple model of the generation of 'spikes' by a single neuron. It is given by the 2-dimensional nonlinear ordinary differential equation (ODE)

$$\begin{aligned}\dot{v} &= v - \frac{v^3}{3} - u + F,\\ \dot{u} &= \epsilon\left(v + A - Bu\right),\end{aligned}$$

where F, A, B and ϵ are constants. The variable $v(t)$ represents the neuron's membrane potential and $u(t)$ represents the recovery current that restores the steady state of the system. F represents an external input (or *forcing*) and ϵ determines the difference in the time scale of $v(t)$ and $u(t)$.

We can convert the ODE into a difference (discrete time) equation using Euler's discretization scheme with time step $h > 0$, namely,

$$\begin{aligned}v_n &= v_{n-1} + h\left(v_{n-1} - \frac{v_{n-1}^3}{3} - u_{n-1} + F\right),\\ u_n &= u_{n-1} + h\epsilon\left(v_{n-1} + A - Bu_{n-1}\right),\end{aligned}$$

where n is discrete time.

We introduce a stochastic perturbation in the second equation in order to make the sequence $\mathbf{x}_n = \begin{bmatrix} v_n \\ u_n \end{bmatrix}$ random. Specifically, we consider the pair of equations

$$v_n = v_{n-1} + h\left(v_{n-1} - \frac{v_{n-1}^3}{3} - u_{n-1} + F\right), \tag{5}$$

$$u_n = u_{n-1} + h\epsilon\left(v_{n-1} + A - Bu_{n-1}\right) + z_n, \tag{6}$$

where $z_n \sim \mathcal{N}(z_n; 0, \sigma_z^2)$. Note that, even if there is no random perturbation in the equation for v_n, this sequence is random as well because it depends on u_n.

We assume observations are collected for the variable v_n alone. In particular, we have

$$y_n = v_n + s_n, \quad n = 1, 2, ..., \tag{7}$$

where $s_n \sim \mathcal{N}(s_n; 0, \sigma_s^2)$ is an i.i.d. noise sequence.

2.2 Exercises

2.2.1 Simulation of the model

Set parameter values $h = 0.01, F = \frac{1}{2}$, $\epsilon = \frac{1}{12}$, $A = \frac{7}{10}$, $B = \frac{8}{10}$, $\sigma_z^2 = \frac{h}{2}$ and $\sigma_s^2 = 4$ and draw $u_0, v_0 \sim \mathcal{U}(0, 1)$. Set $m = 2 \times 10^4$ and simulate the

system described by equations (5), (6) and (7) for $n = 0, 1, 2, ..., m$. Then, for 3 independent simulation runs:

1. Plot v_n versus u_n, $n = 0, 1, \ldots, m$.
2. Plot u_n versus time for $n = 0, 1, \ldots, m$.
3. Plot, on the same axes, the state sequence v_n and the observation sequence y_n versus time $n = 1, \ldots, m$. Use different colours to represent the two signals.

2.2.2 Particle filter

Run a standard PF with $N = 100$ particles for the system described by the prior random distribution $u_0, v_0 \sim \mathcal{U}(0, 1)$ and equations (5), (6) and (7) for $n = 0, 1, 2, \ldots, m$, and $m = 2 \times 10^4$.

Use the PF of Section 1.2.3 as a template. Denote the particles as

$$\bar{\mathbf{x}}_n^i = \begin{bmatrix} \bar{v}_n^i \\ \bar{u}_n^i \end{bmatrix} \quad \text{and} \quad \mathbf{x}_n^i = \begin{bmatrix} v_n^i \\ u_n^i \end{bmatrix},$$

before and after the resampling step, respectively. Note that the procedure for the computation of the importance weights is identical to the one described in Section 1.2.3 if we start with

$$\lambda_n^i = -\frac{1}{2\sigma_s^2}(y_n - v_n^i)^2,$$

for $i = 1, \ldots, N$. The resampling procedure is identical, since it does not depend on the model.

Denote the estimates as

$$\hat{v}_n^N = \sum_{i=1}^{N} \omega_n^i \bar{v}_n^i \quad \text{and} \quad \hat{u}_n^N = \sum_{i=1}^{N} \omega_n^i \bar{u}_n^i.$$

For three independent simulation runs:

- Plot, on the same axes, the true signals v_n versus u_n, and their estimates $\hat{v}_n^N$ versus $\hat{u}_n^N$, for $n = 0, 1, \ldots, m$. Use colors to make the two trajectories clearly visible.
- Plot, on the same axes, the true signal v_n, the observation y_n and the estimate $\hat{v}_n^N$ versus time $n = 0, 1, \ldots, m$. Use colors to make the three signals clearly visible.
- Plot, on the same axes, the true signal u_n and the estimate $\hat{u}_n^N$ versus time $n = 0, 1, \ldots, m$. Use colors to make the two signals clearly visible.

Calculate and display the normalized average square errors over time,

$$\mathcal{E}_v = \frac{\sum_{n=1}^{m}(v_n - \hat{v}_n^N)^2}{\sum_{n=1}^{m} v_n^2} \quad \text{and} \quad \mathcal{E}_v = \frac{\sum_{n=1}^{m}(u_n - \hat{u}_n^N)^2}{\sum_{n=1}^{m} u_n^2}.$$

You may want to try and see the results with larger values of N.

2.2.3 Particle filter with nonlinear observations

Generate nonlinear observations of the form

$$y_n = v_n^2 + s_n,$$

where $s_n \sim \mathcal{N}(s_n; 0, \sigma_s^2)$ and $\sigma_s^2 = 4$, and repeat the experiments of Section 2.2.2. What do you observe?

3 Submission

Prepare a PDF document containing the plots requested in the different exercises and your answers to the questions. You also need to submit the scripts that you have coded to generate the plots in all the exercises.

Put all documents and scripts in a single compressed file (.zip) and upload it using the link in Aula Global.

You can work on this lab in groups of two students. Nevertheless, both students must submit the (same) files to Aula Global and do not forget to include both names in all the files you submit.